

From public PDFs to checked evidence

Retrieval locates source text. Full-page context and exact quotations make that text inspectable.

OFFLINE PREPARATION

Extract source pages

PyMuPDF / public PDFs
Clean text; retain source and page
PDF page numbers are one-based

Make overlapping chunks

1,000 characters / 150 overlap
1,107 Constitution + 42 NCERT
Persist as JSONL with provenance

Build a local index

BAAI/bge-small-en-v1.5
Normalized chunk embeddings
NumPy similarity search

PER QUERY

Build the query

Generation: topic + curated focus
Statements: each complete claim
Direct: stem and all four options

Semantic search

Embed the query
Search the complete loaded index
Select the top 20 candidates

Cross-encoder reranking

Score each query-passage pair
MS MARCO MiniLM-L-6-v2
Keep the top 5 per query

Named Article lookup

Find the actual provision heading
Include continuation and notes
Approved Constitution source only

Restore source context

Reassemble retrieved full pages
Retain qualifications and footnotes
Record added Article pages separately

Quoted evidence packet

Numbered source excerpts
Models select quotation references
Python binds text, source and page

Named references

Ranked selections

Article context adds source pages beside the ranking; it is used for generation and statement checks.
Retrieval scores are ranking signals. A passage hit or an exact quote does not establish an answer.